Messta

SMART DINING OPERATING SYSTEM

Enterprise Platform Documentation

OFFICIAL PLATFORM & OPERATIONS MANUAL

MESSTA User Manual & Operating Blueprint

A complete guide for Students, Mess Owners, Counter Cashiers, Kitchen Managers, and Accountants covering Digital Dining, Dual QR Passes, Smart Kitchen Inventory, Shift Rostering, and P&L Intelligence.

STUDENT PORTAL | COUNTER QR SCANNER | KITCHEN INVENTORY OS | STAFF & CUSTOM ROLES | FINANCE & UNIT ECONOMICS

MESSTA Engineering & Product Team

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Confidential & Operational Manual

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Executive Summary

MESSTA transforms student dining and mess management across Indian campuses. It replaces paper registers, manual cash collection, and untracked kitchen leakage with an automated, cloud-based platform connecting diners directly with kitchen operations.

Every student pass verification triggers automated ingredient deductions in the pantry and updates real-time financial ledgers with zero manual bookkeeping.

Messta End-to-End System Architecture

USER ROLE	PRIMARY ACCESS ROUTE	CORE TOUCHPOINTS	SECURITY & ACCESS LEVEL
Student / Diner	`/mess`, `/membership`	Mess discovery, pass purchase, dynamic QR display, slot booking	User Auth (Student Framework Protected)
Counter Operator	`/owner` (Counter Scanner)	Camera QR scanning, 4-digit PIN verification, token redemption	PIN-based staff audit logging
Kitchen Manager	`/owner/inventory`, `/owner/meals`	Ingredient stock-in, recipe thresholds, daily thali publishing	Inventory & Meal Management permission
Mess Owner / Admin	`/owner/*`	Staff hiring, custom roles, P&L reports, gateway payouts, outlet config	Full Owner Admin Privileges

Key Technical Innovation: Real-Time Event Loop

When a student scans their pass at the counter:
(1) Instant Token Balance Check → **(2)** Order transition to `redeemed` → **(3)** Staff ID & Shift tagged to transaction → **(4)** Raw ingredient COGS deducted in Inventory → **(5)** Live entry posted in Double-Entry Finance Ledger.

Student Portal & Dual QR Architecture

1. Student Discovery & Meal Booking Workflow

1 Find Mess Outlet

Students navigate to `/mess`, browse nearby partner outlets with live distance, meal spread photos, cleanliness ratings, and pricing.

2 Choose Membership or Plate

Purchase a monthly token pass (e.g., 60 Veg tokens @ ₹2,800) or buy a one-off single plate (e.g., Sunday Special Paneer Thali @ ₹110).

3 Generate Secure Pass

Access the pass from `/membership`. A cryptographically signed dynamic QR code displays with time-expiring rolling tokens.

2. Dual QR System Breakdown

ATTRIBUTE	A: MEMBERSHIP TOKEN PASS	B: PER-PLATE SINGLE TICKET
Intended Use	Daily enrolled students with monthly/semester packages	Casual diners, guest plates, out-of-plan special thalis
Pass Lifecycle	Multi-use declining balance (e.g., 58 / 60 Tokens remaining)	Single-use ticket (valid only for specific date & meal slot)
QR Data Payload	`userId`, `subscriptionId`, `messId`, `tokenType: "membership"`, `nonce`	`orderId`, `bookingDate`, `mealType`, `thaliName`, `tokenType: "per_plate"`
Redemption Action	Decrements 1 token from subscription balance; logs redemption date	Marks order status from `active` to `redeemed` immediately
Student Interface	`StudentMembership.jsx` with slot generator button	`MessDetail.jsx` / Profile with instant QR receipt

Slot-Based Anti-Crowding Engine

To eliminate chaotic dining hall queues, the system allocates fixed capacity intervals (e.g., *12:30 PM – 01:00 PM: 35 Seats*). Students choose their preferred time slot when generating their pass. The counter operator's screen displays incoming reservations per slot to guide food warm-up quantities.

🏢 Owner Dashboard & Live Counter Scanner

1. Counter Token Scanner Operations (`/owner`)

The counter scanner is the central operational workstation of the mess hall. It operates seamlessly on mobile phones, tablets, or desktop webcams.

Scanner Security & PIN Verification

- **Staff Sign-in PIN:** Each counter operator enters their unique 4-digit PIN before scanning commences.
- **Shift Accountability:** All redemptions are tagged with `scannedByStaffId` and active shift (Morning Lunch / Evening Dinner).
- **Fraud Prevention:** The scanner rejects already redeemed tokens, expired slots, or mismatched mess branches with distinct audio chimes.

Counter Scanner Visual Signals

- **GREEN FLASH:** Valid pass. Student name, plan category (Veg/Non-Veg), and remaining tokens display instantly.
- **YELLOW ALERT:** Mismatch (e.g., Veg pass holder attempting to redeem a Non-Veg special thali).
- **RED BUZZER:** Duplicate scan attempt or exhausted token pass.

2. Multi-Branch Outlet Management

Messta allows franchise and multi-branch mess owners to manage their entire network from a single unified portal.

FEATURE	HOW IT OPERATES	OWNER ACTION
Branch Switching	Top dropdown switcher allows instant switching between outlets (e.g., North Campus vs South Campus).	Select outlet from top header; all rosters, menus, inventory, and finance switch context.
Add New Outlet	Modal input captures Outlet Name, Locality, Full Address, Contact Phone, and Pure Veg / Non-Veg badge.	Click *"+ Add New Mess / Branch"* in dashboard top bar.
Delete Branch	Allows removing decommissioned outlets with confirmation safeguards.	Click the red *"Delete Branch"* button with password confirmation.

Live Meal Queue Feed

The dashboard features a live redemption feed displaying the last 20 students served, their exact timestamp, redeemed thali, and operator name, ensuring zero discrepancy between counter dishing and cashier records.

🍱 Thali Matrix & Operational Spreads

1. Operational Meal Spreads Configuration (`/owner/meals`)

Unlike rigid restaurant menus, mess operations run in distinct daily production batches. Messta structures menus around **Operational Spreads**:

🌞 Breakfast Spread

Window: 07:30 AM – 10:30 AM
Offerings: Poha, Upma, Idli Sambar, Paratha, Tea/Coffee.
Capacity: Typically rapid rotation (15-min intervals).

🍛 Lunch Spread

Window: 12:00 PM – 03:00 PM
Offerings: Executive Paneer Thali, Standard Veg, Dal Tadka, Rice, Chapati.
Capacity: Peak rush (30-min seat blocks).

🌙 Dinner Spread

Window: 07:30 PM – 10:30 PM
Offerings: Special Non-Veg Chicken Thali, Deluxe Veg, Kheer.
Capacity: Evening student surge.

2. Dynamic Spread Creation (+ Add Spread Feature)

Owners can add custom operational spreads (e.g., *Evening Snacks*, *Sunday Feast*, or *Late Night Study Spread*) with full autonomy over timings and thali publishing.

3. The Crucial Link: Membership Tier Mapping

When creating or publishing a daily thali, the owner maps it to eligible membership tiers:

- **Standard Veg Thali:** Valid for *Lite Veg Pass* & *Executive Veg Pass* (Redeems 1 Token).
- **Special Non-Veg Thali:** Valid strictly for *Non-Veg / Combo Pass* tokens.
- **Out-of-Plan Scenario:** If a Veg Pass student orders a Non-Veg plate, the system detects the mismatch and routes them to direct UPI/card checkout (₹140) instead of deducting a token.

4. Real-Time Capacity Limiting

Each time slot has an explicit seat cap (e.g., Max 40 seats). Once 40 students reserve that window, the slot automatically closes on the student app, directing subsequent diners to the next available 30-minute block and eliminating dining hall congestion.

Plans & Subscriber CRM

1. Publishing Membership Plans (`/owner/memberships`)

Mess owners create subscription packages tailored to student budgets and semester schedules:

PLAN NAME	CATEGORY	TOKENS ALLOCATED	MEALS / DAY	STANDARD PRICE
Monthly Unlimited Veg Pass	VEG	60 Tokens (30 Days × 2 Meals)	2 Meals (Lunch + Dinner)	₹2,800 – ₹3,200
Executive Non-Veg Pass	COMBO	60 Tokens (incl. 8 Special Thalis)	2 Meals + Sunday Specials	₹3,600 – ₹4,200
Student Lite Pass	VEG	30 Tokens (Lunch only)	1 Meal / Day	₹1,600 – ₹1,800
Flexi-Token 20 Pack	FLEXI	20 Tokens (Valid 60 Days)	On-Demand (Any meal)	₹1,200

2. Subscriber CRM Operations

Token Balance Management

Owners can search any enrolled student by name, email, or phone. The subscriber ledger shows remaining tokens, total redemptions, plan validity, and recent meal timestamps.

Manual Adjustment: If a student was absent during college holidays, owners can credit compensation tokens with 1 click.

Pass Lifecycle Controls

- **Pause Subscription:** Temporarily freeze tokens when a student travels home during exam breaks.
- **Resume Subscription:** Reactivate pass immediately upon the student's return.
- **Cancel & Refund:** Prorated token refund calculations for students vacating hostels.

Automated Expiry & Renewal Reminders

When a student reaches their last 4 remaining tokens, the system pushes an automated renewal alert prompting them to top up before the 1st of the month, ensuring unbroken cashflow for the mess owner.

🥦 Kitchen Inventory & Raw Material OS

1. Ingredient Master & Categorization (`/owner/inventory`)

Food service inventory is perishable and fast-moving. Messta organizes kitchen pantry stock into three distinct operational groups:

🥦 Daily Fresh Perishables

Vegetables, onions, tomatoes, milk, paneer, curd, green chilies.
Rotation: Daily or alternate-day mandi restocking.

🌾 Dry Staples & Grains

Basmati rice, wheat flour (atta), toor dal, cooking oil, spices, salt.
Rotation: Weekly/monthly bulk wholesale procurement.

📦 Consumables & Packaging

Takeaway meal boxes, silver foil rolls, napkins, cleaning sanitizers.
Rotation: Bi-weekly restocks.

2. Par Levels & Smart Low-Stock Warnings

Minimum Safety Thresholds (Par Levels)

For every item, owners set a safety minimum (e.g., Minimum 25 kg of Cooking Oil, 40 kg of Rice). When stock dips below this limit, the item badge turns **RED (Low Stock)** on the dashboard, prompting immediate vendor reordering before lunch prep begins.

Stock Inflow Logging

When wholesale deliveries arrive, owners log incoming batches: *Item Name*, *Quantity Received*, *Cost Price*, and *Vendor Name*. The cost automatically flows into the Finance Ledger under `inventory_purchase`.

3. The Automated Outflow Engine (Recipe-Based Consumption)

```
Daily Ingredient Consumption = Σ (Served Thalis × Standard Recipe Proportion) + Spoilage Example (100 Paneer Thalis) = 15kg Rice + 10kg Dal + 6kg Paneer + 200 Rotis Atta
```

Whenever student QR passes are scanned at the counter, the system automatically decrements ingredients from pantry stock based on configured recipe ratios. This eliminates manual stock counting and prevents inventory pilferage.

👨‍🍳 Staff Management & Custom Roles

1. Personnel Directory & Shift Security (`/owner/staff`)

Mess personnel operate in high-intensity shifts. Messta gives owners granular role security, shift accountability, and PIN verification.

Shift Windows & Rostering

- 🌞 **Morning Shift:** 06:30 AM – 03:00 PM (Breakfast prep + Lunch service rush).
- 🌙 **Evening Shift:** 04:30 PM – 10:30 PM (Dinner prep + Service rush + Sanitation).
- ⚡ **Full Day Service:** Supervisory staff managing all operations.

Wage Structure Options

- **Monthly Fixed Salary:** For permanent staff (e.g., Head Chef @ ₹26,000/mo).
- **Daily Wage Rate:** For helpers and cleaners (e.g., ₹450 – ₹600 / day).
- **Payroll Auto-Sync:** Monthly wage totals feed directly into the Finance Module as labor overhead.

2. Custom Role Creator (+ Add Role Feature)

Beyond standard system roles (Head Chef, Counter Operator, Delivery Partner), owners can create custom roles tailored to their outlet:

CUSTOM ROLE	ICON	TYPICAL OPERATIONAL RESPONSIBILITIES	DEFAULT PERMISSIONS ENABLED
Front Desk / Cashier	🛎️	Student check-ins, cash token top-ups, counter inquiries	QR Scanner, Staff Check-in
Table & Floor Service	🍽️	Dining hall seating, table clearance, drinking water refilling	View Only (No system modifications)
Storekeeper & Pantry	📦	Dry ration intake, vegetable weigh-ins, inventory audits	Kitchen Inventory Management
Dishwashing & Sanitation	🧼	Kitchen utensil cleaning, dish sanitization, waste disposal	View Only (Daily wage tracking)

Granular Permission Matrix

Owners toggle specific checkboxes per role: `canScanQR`, `canManageInventory`, `canManageMeals`, `canViewFinance`, and `canManageStaff`. Counter staff cannot view financial profits, preventing sensitive data exposure.

💰 Finance & P&L Intelligence Engine

1. The Double-Entry Automated Ledger (`/owner/finance`)

The finance engine unites all operational streams into an automated double-entry ledger:

```
Net Profit = Net Revenue - (Kitchen COGS + Staff Payroll + LPG & Power + Venue Rent + Repairs)
Target Standard: Gross Margin ≥ 50% | Net Profit Margin = 26% - 34% (Healthy)
```

Inflow Channels (Revenue)

- **Subscriptions:** ~80% of revenue from recurring student passes.
- **Per-Plate Meals:** ~16% from walk-in diners and special dishes.
- **Counter Add-ons:** ~4% from cash/UPI sweets, extra curd, and drinks.
- **Net Inflow:** Programmatically deducts 2% + GST gateway charges.

Outflow Channels (Expenses)

- **Food COGS:** Real-time consumption cost of raw ingredients.
- **Staff Payroll:** Monthly salaries + daily shift wages.
- **LPG Gas & Utilities:** 19kg commercial cylinders, electricity bills.
- **Venue Rent:** Monthly mess hall lease.

2. Per-Thali Unit Economics & Contribution Margins

THALI DISH NAME	SELLING PRICE	RAW MATERIAL COGS	FUEL & LABOR SHARE	GROSS MARGIN %	CONTRIBUTION MARGIN	PERFORMANCE RATING
Executive Paneer Thali	₹110	₹44	₹18 (₹6 gas + ₹12 labor)	**60.0%**	**+₹48**	HIGH MARGIN STAR ⭐
Standard Veg Thali	₹85	₹32	₹18 (₹6 gas + ₹12 labor)	**62.4%**	**+₹35**	CORE STAPLE
Special Non-Veg Chicken Thali	₹145	₹68	₹20 (₹7 gas + ₹13 labor)	**53.1%**	**+₹57**	HIGH REVENUE
Student Lite Thali	₹65	₹24	₹15 (₹5 gas + ₹10 labor)	**63.1%**	**+₹26**	BUDGET STAPLE

3. Break-Even Daily Plate Calculator

Fixed Monthly Overhead: ₹48,000 (Rent ₹22k + Base Salaries ₹20k + Electricity ₹6k)
Average Contribution Margin: ₹36 per plate
Daily Break-Even Target: 45 Plates / Day (1,350 plates/month) to cover all fixed expenses. Every plate served past 45 represents pure net profit.

Quick Reference Cheatsheet & Troubleshooting

Daily Operations Checklist for Mess Owners

TIME	MODULE	REQUIRED ACTION
06:30 AM	Kitchen Inventory	Inspect vegetable delivery, verify par levels for cooking oil and rice, log Mandi purchase.
07:00 AM	Thali Matrix	Publish Today's Breakfast & Lunch Thalis, set capacity slots (e.g. 12:00–12:30, 12:30–1:00).
11:45 AM	Counter Scanner	Cashier enters 4-digit PIN, verifies webcam scanner focus, reviews incoming slot reservations.
03:15 PM	Staff Roster	Mark Morning Shift completion; review redemption audit count vs lunch headcount.
05:30 PM	Thali Matrix	Confirm Evening Dinner Spread (Paneer/Chicken Specials); update night capacities.
10:30 PM	Finance Ledger	Quick-add any daily cash expenses (LPG refills, tanker bills); review Day Net Margin %.

Frequently Asked Questions & Troubleshooting

Q: A student's QR code won't scan at the counter. What should the cashier do?

A: 1) Ensure the student's phone screen brightness is set to high. 2) If the camera is blurred, the operator can manually enter the student's 6-character Pass Code (e.g., `PASS-7A8B`) into the counter fallback box to redeem 1 token instantly.

Q: How do Razorpay / Stripe payouts reach the owner's bank account?

A: Digital transactions collect in payment gateway escrow and batch-settle on a T+1 or T+2 business day schedule into the owner's registered bank account. View all cleared deposits and UTR reference numbers under `/owner/finance → Gateway Settlements`.

Q: Can an owner manage two different messes with separate menus and staff?

A: Yes! Use the top-left branch picker dropdown to toggle between outlets. Each mess maintains its own independent inventory, staff roster, meal spreads, and P&L financial ledgers.

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